UNIT-V

FOURIER SERIES AND FOURIER TRANSFORMS

Topic Learning Objectives:

Upon Completion of this unit, students will be able to:

- Understand the periodic phenomena
- Develop the knowledge of periodic functions as a Fourier series subject to Dirichlet's conditions and derive the Fourier series using Euler's formulae.
- Expand a function in complex form of Fourier series,
- Represent continuous-time periodic signals using Fourier series
- Find sine and cosine series
- Understand the Fourier integral theorem.
- Apply complex Fourier transform, Fourier sine and cosine transforms.
- Demonstrate - the properties of transforms.

FOURIER SERIES

Introduction:

French mathematician and physicist Jacques Fourier for the first time used Fourier series in his work in which he developed the theory of heat conduction. These series had a deep influence in the further development of mathematics and mathematical physics. In many engineering problems especially in the study of periodic phenomena in the conduction of heat, electro-dynamics and acoustics it is necessary to express a function in a series of sines and cosines. Fourier series are infinite series designed to represent general periodic functions in terms of cosines and sines. Fourier series are very important to engineers and physicist as they allow the solution of ODEs in connection with forced oscillations and the approximation of periodic functions. Fourier series are more universal than the Taylor series in calculus as many *discontinuous* periodic functions in applications that do not have Taylor series expansions can be developed in Fourier series.

"JEAN-BAPTISTE JOSEPH FOURIER (1768–1830), French physicist and mathematician, lived and taught in Paris, accompanied Napoléon in the Egyptian War and was later made

prefect of Grenoble. The beginnings on Fourier series can be found in works by Euler and by Daniel Bernoulli, but it was Fourier who employed them in a systematic and general manner in his main work, Théorie analytique de la chaleur (Analytic Theory of Heat, Paris, 1822), in which he developed the theory of heat conduction making these series a most important tool in applied mathematics."

Periodic function

Definition: A general periodic function $f(x)$ of period T satisfies the condition

$$f(x+T) = f(x).$$

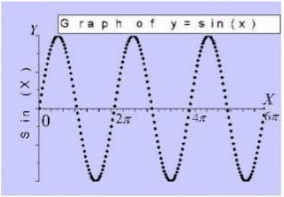
Here $f(x)$ is a real-valued function and $T > 0$ is a positive real number.
As a consequence, it follows that

$$f(x) = f(x+T) = f(x+2T) = f(x+3T) = \ldots\ldots = f(x+nT).$$

Thus,

$$f(x) = f(x+nT),\ n=1,2,3,\ldots\ldots$$

The function $f(x) = \sin x$ is periodic of period 2π since $\sin(x+2n\pi) = \sin x$, $n=1, 2, 3,\ldots\ldots$
The graph of the function is shown below

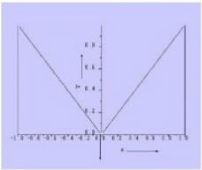


Note that the graph of the function between 0 and 2π is the same as that between 2π and 4π and so on. It may be verified that a linear combination of periodic functions is also periodic.
Examples: $\cos x$, $\sin x$, $\sec x$, $\operatorname{cosec} x$, are periodic function with period 2π.
$\tan x$, $\cot x$ are periodic with period π .

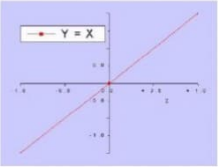
Even and Odd functions:

Definition: A function $y = f(x)$ is said to be <u>even</u>, if $f(-x) = f(x)$. The graph of the even function is always symmetrical about the y-axis.
A function $y= f(x)$ is said to be <u>odd</u>, if $f(-x) = - f(x)$. The graph of the odd function is always symmetrical about the origin.

Examples: $f(x) = |x|$ is an even function.
$f(x) = x$ is an odd function.



Graph of $f(x) = |x|$



Graph of $f(x) = x$

Note that the graph of $f(x) = |x|$ is symmetrical about the y-axis and the graph of $f(x) = x$ is symmetrical about the origin.

Note:

1. If $f(x)$ is even and $g(x)$ is odd, then
   - $h(x) = f(x)\, g(x)$ is odd
   - $h(x) = f(x)\, f(x)$ is even
   - $h(x) = g(x)\, g(x)$ is even

multiplication → odd
squaring odd or even → even

Examples:

1. $h(x) = x^2 \cos x$ is even, since both x^2 and $\cos x$ are even functions.
2. $h(x) = x \sin x$ is even, since x and $\sin x$ are odd functions.
3. $h(x) = x^2 \sin x$ is odd, since x^2 is even and $\sin x$ is odd.

Dirichlet's conditions:

Consider a real-valued function $f(x)$ which obeys the following conditions called Dirichlet's conditions :

1. $f(x)$ is defined in an interval $(a, a+2l)$, and $f(x+2l) = f(x)$ so that $f(x)$ is a periodic function of period $2l$.
2. $f(x)$ is continuous or has only finite number of discontinuities in the interval $(a, a+2l)$.
3. $f(x)$ has no or only a finite number of maxima or minima in the interval $(a, a+2l)$.

Here

$$a_0 = \frac{1}{l}\int_a^{a+2l} f(x)\,dx, \tag{1}$$

$$a_n = \frac{1}{l}\int_a^{a+2l} f(x)\cos\left(\frac{n\pi}{l}\right)x\,dx, \quad n = 1,2,3,\ldots.. \tag{2}$$

$$b_n = \frac{1}{l}\int_a^{a+2l} f(x)\sin\left(\frac{n\pi}{l}\right)x\,dx, \quad n = 1,2,3,\ldots\ldots. \tag{3}$$

Then the infinite series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}\right)x + b_n \sin\left(\frac{n\pi}{l}\right)x, \tag{4}$$

is called the Fourier series of $f(x)$ in the interval $(a, a+2l)$. Also, the real numbers $a_0, a_1, a_2, \ldots a_n$, and $b_1, b_2, \ldots b_n$ are called the Fourier coefficients of $f(x)$. The formulae (1), (2) and (3) are called **Euler's formulae**.

It can be proved that the sum of the series (4) is $f(x)$ if $f(x)$ is continuous at x. Thus

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}\right)x + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{l}\right)x. \tag{5}$$

Suppose $f(x)$ is discontinuous at x, then the sum of the series (4) would be

$$\frac{1}{2}\left[f(x^+) + f(x^-)\right],$$

where $f(x^+)$ and $f(x^-)$ are the values of $f(x)$ immediately to the right and to the left of $f(x)$ respectively.

Particular Cases:

Case (i)

Suppose $a = 0$. Then $f(x)$ is defined over the interval $(0,2l)$. Formulae (1), (2), (3) reduce to

$$a_0 = \frac{1}{l}\int_0^{2l} f(x)dx, \quad a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\left(\frac{n\pi}{l}\right)xdx,$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\left(\frac{n\pi}{l}\right)xdx \;.\; n = 1,2,.....\infty, \tag{6}$$

Then the right-hand side of (5) is the Fourier expansion of $f(x)$ over the interval $(0,2l)$.

If $l = \pi$, then $f(x)$ is defined over the interval $(0, 2\pi)$. Formulae (6) reduces to

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx,$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nxdx, \tag{7}$$

$$n = 1,2,-----\infty,$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nxdx,$$

$$n = 1,2,-----\infty.$$

Also, in this case, (5) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx. \tag{8}$$

Case (ii)

Suppose $a = -l$. Then $f(x)$ is defined over the interval $(-l, l)$. Formulae (1), (2), (3) reduce to

$$a_0 = \frac{1}{l}\int_{-l}^{l} f(x)dx,$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\left(\frac{n\pi}{l}\right)xdx, \tag{9}$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\left(\frac{n\pi}{l}\right)xdx.$$

Then the right-hand side of (5) is the Fourier expansion of $f(x)$ over the interval $(-l, l)$.

If $l = \pi$, then $f(x)$ is defined over the interval $(-\pi, \pi)$. Formulae (9) reduce to

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx,$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nxdx, \qquad n = 1,2,.....\infty \tag{10}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nxdx, \qquad n = 1,2,.....\infty.$$

Putting $l = \pi$ in (5),

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx.}$$

Expressing $\cos nx$ and $\sin nx$ in exponential form, complex form of Fourier series is obtained.

Complex form of Fourier Series:

The Complex Fourier Series of $f(x)$ is defined to be

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx/l}, \text{ where } c_n = \frac{1}{2l}\int_a^{a+2l} f(x) e^{-inx/l}\, dx.$$

In particular $\boxed{\text{when } a = -l,\ l = \pi,}$

$$\boxed{f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \text{ where } c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\, dx.}$$

Here $\boxed{c_n = \frac{1}{2}(a_n - ib_n),\ c_{-n} = \frac{1}{2}(a_n + ib_n) \text{ and } c_0 = \frac{1}{2}a_0,}$ $n > 0$,

$$\Rightarrow \boxed{a_0 = 2c_0,\ \ a_n = c_n + c_{-n},\ b_n = i(c_n - c_{-n}).}$$

Examples:

1. Find the complex Fourier series of $f(x) = x, -\pi < x < \pi$. Hence deduce the trigonometric series.

Solution:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \text{ where } c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\, dx$$

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} x e^{-inx}\, dx$$

$$= -\frac{(-1)^n}{in}, n \neq 0$$

$$\text{for } n = 0,\ c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\, dx = 0$$

$$\text{Thus, } f(x) = \sum_{n=-\infty, n\neq 0}^{\infty} \frac{(-1)^{n+1}}{in} e^{inx}$$

$$\boxed{e^{-inx} = \cos(nx) - i\sin(nx)}$$

also,
any odd func^n (like cos) integrated from $(-a, a)$
$= 0$.

Corresponding trigonometric series:

$$a_0 = 2c_0 = 0,\ a_n = c_n + c_{-n} = 0,\ b_n = i(c_n - c_{-n}) = 2\frac{(-1)^{n+1}}{n},$$

Required trigonometric series is

$$f(x) = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx.$$

2. Obtain the complex Fourier series for the function $(x) = \begin{cases} 0, & for\ -\pi < x < 0 \\ \sin x, & for\ \ 0 < x < \pi \end{cases}$.

Hence deduce the trigonometric series.

Solution: Given $f(x) = \begin{cases} 0, & for\ -\pi < x < 0 \\ \sin x, & for\ \ 0 < x < \pi \end{cases}$.

The complex form of Fourier series is given by:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

where $c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{inx}\,dx,$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\sin x\,dx,$$

$$= \frac{1}{2\pi}\left[\frac{e^{-inx}}{(-in)^2+1^2}\{-in\sin x - 1.\cos x\}\right]_0^{\pi},$$

$$= \frac{1}{2\pi}\left[\frac{e^{-inx}}{1-n^2}\{-in\sin x - 1.\cos x\}\right]_0^{\pi}, \quad \text{for } n \neq \pm 1$$

$$= \frac{1}{2\pi(1-n^2)}\left[-\{e^{-inx}\cos\pi - e^0\cos 0\}\right],$$

$$c_n = \frac{1}{2\pi(1-n^2)}[(-1)^n + 1],$$

$$= -\frac{[(-1)^n+1]}{2\pi(n^2-1)}.$$

For n =1, $\quad c_1 = \frac{1}{2\pi}\int_0^{\pi} e^{-inx}\sin x\,dx = \frac{-i}{4}$.

For n = -1, $\quad c_{-1} = \frac{1}{2\pi}\int_0^{\pi} e^{ix}\sin x\,dx = \frac{i}{4}$.

Substituting the above values in (1)

$$f(x) = c_0 + c_{1e^{ix}+\sum_{n=2}^{\infty} c_n e^{inx}+c_{-1}e^{-ix}+\sum_{n=-\infty}^{-2} c_n e^{inx},}$$

$$f(x) = \frac{1}{\pi} - \frac{i}{4}e^{ix} + \frac{i}{4}e^{-ix} - \frac{1}{2\pi}\left\{\sum_{n=2}^{\infty}\frac{1+(-1)^n}{n^2-1}e^{inx} + \sum_{n=-\infty}^{-2}\frac{1+(-1)^n}{n^2-1}e^{inx}\right\}.$$

Corresponding trigonometric series:

$$a_0 = 2c_0 = \frac{2}{\pi},\ a_n = c_n + c_{-n} = \frac{-\left[1+(-1)^n\right]}{\pi(n^2-1)}, n \geq 2,\ a_1 = c_1 + c_{-1} = 0$$

$$b_n = i(c_n - c_{-n}) = 0,\ n \geq 2,\ b_1 = i(c_1 - c_{-1}) = \frac{1}{2},$$

$$f(x) = \frac{1}{\pi} + \frac{\sin x}{2} - \frac{2}{\pi}\sum_{m=1}^{\infty}\frac{\cos 2mx}{4m^2-1}.$$

3. Find the complex form of the Fourier Series of $f(x) = e^x$ in $-1 \leq x \leq 1$. Hence deduce the trigonometric series.

Solution: The complex form of Fourier series is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \qquad (l=1)$$

while $\quad c_n = \frac{1}{2}\int_{-1}^{1} e^{-x}e^{in\pi x}dx,$

$$=\frac{1}{2}\int_{-1}^{1} e^{-(1+in\pi)x},$$
$$=\frac{1}{2}\left[\frac{e^{-1(1+in\pi)x}}{-(in\pi)}\right]_{-1}^{1},$$
$$=\frac{1}{2}\left[\frac{e^{1}e^{in\pi}-e^{-1}e^{-in\pi}}{(1+in\pi)}\cdot\frac{1-in\pi}{1+in\pi}\right]=\frac{(-1)^n(1-in\pi)}{1^2+n^2\pi^2}\sinh 1,$$
$$e^{-x}=\sum_{n=-\infty}^{\infty}\frac{(-1)^n(1-in\pi)}{1^2+n^2\pi^2}\sinh 1\ e^{in\pi x}.$$

Exercise:
Obtain the Complex Fourier Series expansion of the following functions:

1. $f(x)=\begin{cases}0, & 0<x<l\\ a, & 0<x<2l\end{cases}$ over the interval $(0, 2l)$. Hence deduce the trigonometric series.

Answer: $f(x)=\frac{a}{2}-\frac{ia}{2\pi}\sum_{n=-\infty}^{\infty}\frac{\left[1-(-1)^n\right]}{n}e^{i\left(\frac{n\pi}{l}\right)x}, n\neq 0.$

$$f(x)=\frac{a}{2}-\frac{a}{\pi}\sum_{n=1}^{\infty}\frac{\left[1-(-1)^n\right]}{n}\sin\left(\frac{n\pi}{l}\right)x,\quad n\neq 0.$$

2. $f(x)=x\sin x$ over the interval $(0,2\pi)$.Hence deduce the corresponding trigonometric series.

Answer: $f(x)=-1-\left(\frac{1}{4}+\frac{\pi}{2}i\right)e^{ix}-\left(\frac{1}{4}-\frac{\pi}{2}i\right)e^{-ix}+\sum_{n=-\infty}^{-2}\frac{1}{n^2-1}e^{inx}+\sum_{n=2}^{\infty}\frac{1}{n^2-1}e^{inx},$

$$f(x)=-1-\frac{1}{2}\cos x+\pi\sin x+2\sum_{n=2}^{\infty}\frac{1}{n^2-1}\cos nx.$$

3. $f(x)=\begin{cases}-k, & -\pi<x<0\\ k, & 0<x<\pi\end{cases}$ where 'k' is a real constant over the interval $(-\pi,\pi)$.

Answer: $f(x)=-\frac{ik}{\pi}\sum_{n=-\infty}^{\infty}\frac{\left[1-(-1)^n\right]}{n}e^{inx}, n\neq 0,$

$$f(x)=\frac{2k}{\pi}\sum_{n=1}^{\infty}\frac{\left[1-(-1)^n\right]}{n}\sin nx.$$

Examples on time periodic signals:

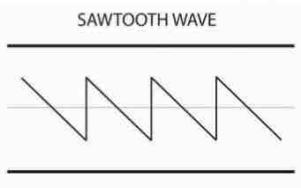
Special wave forms:
Special wave forms can be classified as:
1. Square wave
2. Half wave rectifier
3. Saw tooth rectifier
4. Triangular wave

Saw tooth wave form:
1. Express $f(x) = x/2$ as a Fourier series in the interval $-\pi < x < \pi$.
Solution: Since $f(-x)= - x/2 = f(x)$.

odd: $a_0 = a_n = 0$
$b_n = \frac{2}{l}\int_0 f(x)\sin(nx)\,dx$
even : $b_n = 0$

Therefore, $f(x)$ is an odd function and
$f(x)=\sum_{n=1}^{\infty} b_n \sin nx,$
$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx,$
$b_n = \frac{2}{\pi}\int_0^{\pi} \frac{x}{2}\sin nx\,dx,$
$b_n = \frac{1}{\pi}\left[x\left(-\frac{\cos nx}{n}\right) - \left(-\frac{\sin nx}{n^2}\right)\right] = \frac{\cos n\pi}{n},$
$b_1 = 1, b_2 = -\frac{1}{2}, b_3 = \frac{1}{3}, b_4 = -\frac{1}{4}$ etc..... .
Hence the series is $\frac{x}{2} = \sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \cdots$.

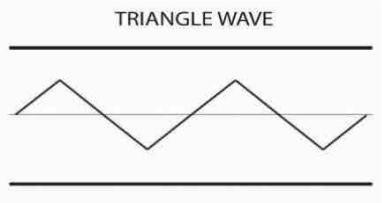
The above series represents the discontinuous function called saw-tooth waveform shown in the figure below.



Triangular waveform:

2. Obtain Fourier series for the function $f(x)$ given by $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi \le x \le 0 \\ 1 - \frac{2x}{\pi}, & 0 \le x \le \pi \end{cases}$.

Solution: Since $f(-x)=1 - \frac{2x}{\pi}$ in $(-\pi,0) = f(x)$ in $(0,\pi)$,
$f(x) = 1+\frac{2x}{\pi}$ in $(0,\pi) = f(x)$ in$(-\pi,0)$,
therefore $f(x)$ is an even function in$(-\pi,\pi)$ and the graph is symmetrical about the y-axis, the required series is
$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx),$
where, $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx = \frac{2}{\pi}\int_0^{\pi}\left(1 - \frac{2x}{\pi}\right)dx = 0,$
$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi}\left(1 - \frac{2x}{\pi}\right)\cos nx\,dx,$
$= \frac{4}{n^2\pi^2}[1 - (-1)^n],$
$a_1 = \frac{8}{\pi^2}, a_3 = \frac{8}{3^2\pi^2}, a_5 = \frac{8}{5^2\pi^2}, \cdots$
$a_2 = a_4 = a_6 = \cdots = 0.$
Thus substituting the values of a's in the required series,
$f(x) = \frac{8}{\pi^2}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \cdots\right]$ is the required Fourier series.



Square wave form:

3. Find the Fourier series of the periodic function $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$.

Functions of this kind occur as external forces acting on mechanical systems, electromotive forces in electric circuits, etc.

Solution: The Fourier series of $f(x)$ defined in$(-\pi, \pi)$ having a period of 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \ldots.,$$

$$a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -k dx + \int_{0}^{\pi} k dx\right\},$$

$$= \frac{k}{\pi}\{0\} = 0,$$

$$\frac{a_0}{2} = 0,$$

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -k \cos nx\, dx + \int_{0}^{\pi} k \cos nx\, dx\right\},$$

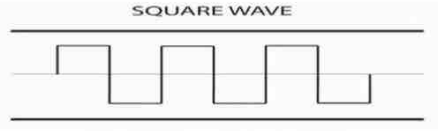
$$a_n = 0,$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -k \sin nx\, dx + \int_{0}^{\pi} k \sin nx\, dx\right\},$$

$$b_n = \frac{2k}{\pi n}\{1 - (-1)^n\}.$$

The required Fourier series is given by

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{\pi n}\{1 - (-1)^n\} \sin nx.$$



4. An alternating current after passing through a rectifier has the form

$I = \begin{cases} -I_0 \sin\theta, & 0 < \theta < \pi \\ 0, & \pi < \theta < 2\pi \end{cases}$ where I_0 is the max current.

Express I as Fourier series in $(0, 2\pi)$.

Solution: The Fourier series of period 2π is given by

$$I = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\theta + \sum_{n=1}^{\infty} b_n \sin n\theta,$$

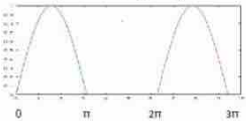
$$a_0 = \frac{I_0}{\pi}\int_0^{\pi} \sin\theta d\theta,\ a_n = \frac{I_0}{\pi}\int_0^{\pi} \sin\theta \cos n\theta\ d\theta,\ b_n = \frac{I_0}{\pi}\int_0^{\pi} \sin n\theta\ d\theta,$$

$$a_0 = \frac{2I_0}{\pi},$$

$$a_n = \frac{-I_0}{\pi(n^2-1)}\{1+(-1)^n\}\ if\ n \neq 1, a_1 = 0, b_n = 0\ if\ n \neq 1 \text{ and } b_1 = \frac{I_0}{2}.$$

Substituting in (1);

$$I = \frac{I_0}{\pi} + \sum_{n=2}^{\infty} \frac{-1}{\pi(n^2-1)}\{1+(-1)^n\}\cos n\theta + \frac{I_0}{2}\sin\theta.$$



Example for Fourier expansion of continuous functions:

1. Obtain the Fourier expansion of $f(x) = \frac{1}{2}(\pi - x)$ in $-\pi < x < \pi$.

Solution:

$a = -\pi.$

$\therefore a + 2l = \pi$

$-\pi + 2l = \pi$

$\boxed{\therefore l = \pi}$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1}{2}(\pi - x)dx,$$

$$= \frac{1}{2\pi}\left[\pi x - \frac{x^2}{2}\right]_{-\pi}^{\pi} = \pi,$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx = \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1}{2}(\pi - x)\cos nx dx,$$

Integration by parts,

$$a_n = \frac{1}{2\pi}\left[(\pi - x)\frac{\sin nx}{n} - (-1)\left(\frac{-\cos nx}{n^2}\right)\right]_{-\pi}^{\pi},$$

$$= \frac{1}{2\pi}[0] = 0,$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1}{2}(\pi - x)\sin nx dx,$$

$$= \frac{1}{2\pi}\left[(\pi - x)\frac{-\cos nx}{n} - (-1)\left(\frac{-\sin nx}{n^2}\right)\right]_{-\pi}^{\pi},$$

$$= \frac{(-1)^n}{n}.$$

Using the values of a_0 , a_n and b_n in the Fourier expansion

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

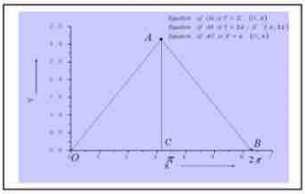
$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin nx.$$

This is the required Fourier expansion of the given function.

2. Obtain the Fourier expansion of

$f(x)=\begin{cases} x, & 0\le x\le \pi \\ 2\pi - x, & \pi \le x \le 2\pi \end{cases}$. Deduce that $\dfrac{\pi^2}{8}=1+\dfrac{1}{3^2}+\dfrac{1}{5^2}+.....$

Solution: The graph of $f(x)$ is shown below.



Here OA represents the line $f(x) = x$, AB represents the line $f(x) = (2\pi\text{-}x)$ and AC represents the line $x = \pi$. The graph is symmetrical about the line AC, which in turn is parallel to y-axis. Hence the function $f(x)$ is an even function.

Here, $a_0=\dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} f(x)dx=\dfrac{2}{\pi}\int_0^{\pi} f(x)dx,$

$$=\frac{2}{\pi}\int_0^{\pi} x dx=\pi$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx=\frac{2}{\pi}\int_0^{\pi} f(x)\cos nx dx$$

Since $f(x)\cos nx$ is even.

$$=\frac{2}{\pi}\int_0^{\pi} x\cos nx dx \qquad =\frac{2}{\pi}\left[x\left(\frac{\sin nx}{n}\right)-1\left(\frac{-\cos nx}{n^2}\right)\right]_0^{\pi}=\frac{2}{n^2\pi}\left[(-1)^n-1\right].$$

Also, $b_n=\dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} f(x)\sin nx dx=0$, since $f(x)\sin nx$ is odd.

Thus the Fourier series of $f(x)$ is $f(x)=\dfrac{\pi}{2}+\dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}\left[(-1)^n-1\right]\cos nx.$

For $x = \pi$ a point of continuity,

$$f(\pi)=\frac{\pi}{2}+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^2}\left[(-1)^n-1\right]\cos n\pi,$$

or $\quad \pi=\dfrac{\pi}{2}+\dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{-2\cos(2n-1)\pi}{(2n-1)^2}.$

Thus, $\dfrac{\pi^2}{8}=\displaystyle\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}$ or $\dfrac{\pi^2}{8}=1+\dfrac{1}{3^2}+\dfrac{1}{5^2}+.....$.

This is the series as required.

Example for discontinuous functions:

3.Obtain the Fourier expansion of

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi. \end{cases}$$

Deduce that $\dfrac{\pi^2}{8} = 1 + \dfrac{1}{3^2} + \dfrac{1}{5^2} + \ldots\ldots$.

Solution: Here, $a_0 = \dfrac{1}{\pi}\left[\int_{-\pi}^{0} -\pi dx + \int_{0}^{\pi} x dx\right] = -\dfrac{\pi}{2}$,

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi \cos nx dx + \int_{0}^{\pi} x \cos nx dx\right],$$

$$= \frac{1}{n^2\pi}\left[(-1)^n - 1\right],$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi \sin nx dx + \int_{0}^{\pi} x \sin nx dx\right],$$

$$= \frac{1}{n}\left[1 - 2(-1)^n\right].$$

Fourier series is $f(x) = \dfrac{-\pi}{4} - \dfrac{1}{\pi}\sum_{n=1}^{\infty}\dfrac{1}{n^2}\left[(-1)^n - 1\right]\cos nx + \sum_{n=1}^{\infty}\dfrac{\left[1 - 2(-1)^n\right]}{n}\sin nx$.

Note, that the point $x = 0$ is a point of discontinuity of $f(x)$.

Here $f(x^+) = 0$, $f(x^-) = -\pi$ at $x = 0$. Hence $\dfrac{1}{2}[f(x^+) + f(x^-)] = \dfrac{1}{2}(0 - \pi) = \dfrac{-\pi}{2}$.

4. Obtain the Fourier expansion of $f(x) = e^{-ax}$ in the interval $(-\pi, \pi)$. Deduce that

$$\pi \operatorname{cosech} \pi = 2\sum_{n=2}^{\infty}\frac{(-1)^n}{n^2+1}.$$

Solution: Here,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{-ax} dx = \frac{1}{\pi}\left[\frac{e^{-ax}}{-a}\right]_{-\pi}^{\pi} = \frac{e^{a\pi} - e^{-a\pi}}{a\pi} = \frac{2\sinh a\pi}{a\pi},$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{-ax}\cos nx dx$$

$$a_n = \frac{1}{\pi}\left[\frac{e^{-ax}}{a^2+n^2}\{-a\cos nx + n\sin nx\}\right]_{-\pi}^{\pi} = \frac{2a}{\pi}\left[\frac{(-1)^n \sinh a\pi}{a^2+n^2}\right]$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{-ax}\sin nx dx$$

$$= \frac{1}{\pi}\left[\frac{e^{-ax}}{a^2+n^2}\{-a\sin nx - n\cos nx\}\right]_{-\pi}^{\pi}$$

$$= \frac{2n}{\pi}\left[\frac{(-1)^n \sinh a\pi}{a^2+n^2}\right].$$

Thus, $f(x) = \dfrac{\sinh a\pi}{a\pi} + \dfrac{2a\sinh a\pi}{\pi}\sum_{n=1}^{\infty}\dfrac{(-1)^n}{a^2+n^2}\cos nx + \dfrac{2}{\pi}\sinh a\pi\sum_{n=1}^{\infty}\dfrac{n(-1)^n}{a^2+n^2}\sin nx.$

For $x = 0$, $a = 1$, the series reduces to

$$f(0)=1 = \frac{\sinh \pi}{\pi} + \frac{2\sinh \pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+1}$$

or

$$1 = \frac{\sinh \pi}{\pi} + \frac{2\sinh \pi}{\pi}\left[-\frac{1}{2} + \sum_{n=2}^{\infty}\frac{(-1)^n}{n^2+1}\right]$$

or

$$1 = \frac{2\sinh \pi}{\pi}\sum_{n=2}^{\infty}\frac{(-1)^n}{n^2+1}$$

Thus, $\pi\cos ech\pi = 2\sum_{n=2}^{\infty}\dfrac{(-1)^n}{n^2+1}.$ This is the desired deduction.

Example for even and odd functions:

5. Obtain the Fourier expansion of $f(x) = x^2$ over the interval $(-\pi, \pi)$. Deduce that $\dfrac{\pi^2}{6} = 1 + \dfrac{1}{2^2} + \dfrac{1}{3^2} + \ldots\ldots + \infty$.

Solution: The function $f(x)$ is even. Hence

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx,$$

$$= \frac{2}{\pi}\int_{0}^{\pi} x^2 dx = \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi},$$

$$a_0 = \frac{2\pi^2}{3}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx,$$

$$= \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx dx, \quad \text{since } f(x)\cos nx \text{ is even}$$

$$= \frac{2}{\pi}\int_{0}^{\pi} x^2 \cos nx dx.,$$

Integrating by parts,

$$a_n = \frac{2}{\pi}\left[x^2\left(\frac{\sin nx}{n}\right) - 2x\left(\frac{-\cos nx}{n^2}\right) + 2\left(\frac{-\sin nx}{n^3}\right)\right]_0^{\pi} = \frac{4(-1)^n}{n^2}.$$

Also, $b_n = \dfrac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx = 0,$ since $f(x)\sin nx$ is odd.

$$f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2}, \qquad (a)$$

$$\pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{1}{n^2}, \qquad \text{(putting } x = \pi \text{ in}(a))$$

$$\sum_{1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Hence, $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + ...$.

Example for arbitrary interval

6. Obtain the Fourier series of $f(x) = 1 - x^2$ over the interval (-1, 1).
Solution: The given function is even as $f(-x) = f(x)$. Also period of $f(x)$ is 1-(-1) = 2.

Here
$$a_0 = \frac{1}{1}\int_{-1}^{1} f(x)dx = 2\int_{0}^{1} f(x)dx,$$

$$= 2\int_{0}^{1} (1 - x^2)dx = 2\left[x - \frac{x^3}{3}\right]_0^1,$$

$$= \frac{4}{3},$$

$$a_n = \frac{1}{1}\int_{-1}^{1} f(x)\cos(n\pi x)dx,$$

$$= 2\int_{0}^{1} f(x)\cos(n\pi x)dx, \qquad \text{as } f(x) \cos(n\pi x) \text{ is even}$$

$$= 2\int_{0}^{1} (1 - x^2)\ \cos(n\pi x)dx.$$

Integrating by parts,

$$a_n = 2\left[(1 - x^2)\left(\frac{\sin n\pi x}{n\pi}\right) - (-2x)\left(\frac{-\cos n\pi x}{(n\pi)^2}\right) + (-2)\left(\frac{-\sin n\pi x}{(n\pi)^3}\right)\right]_0^1,$$

$$= \frac{4(-1)^{n+1}}{n^2\pi^2},$$

$$b_n = \frac{1}{1}\int_{-1}^{1} f(x)\sin(n\pi x)dx = 0, \quad \text{since } f(x) \sin(n\pi x) \text{ is odd.}$$

The Fourier series of $f(x)$ is

$$f(x) = \frac{2}{3} + \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}\cos(n\pi x)$$

7. Obtain the Fourier expansion of

$$f(x) = \begin{cases} 1+\dfrac{4x}{3} & in \quad -\dfrac{3}{2} < x \le 0 \\ 1-\dfrac{4x}{3} & in \quad 0 \le x < \dfrac{3}{2} \end{cases}.$$

Deduce that $\dfrac{\pi^2}{8} = 1 + \dfrac{1}{3^2} + \dfrac{1}{5^2} + \ldots\ldots$.

Solution: The period of $f(x)$ is $\dfrac{3}{2} - \left(\dfrac{-3}{2}\right) = 3.$

Also $f(-x) = f(x)$. Hence $f(x)$ is even.

$$a_0 = \frac{1}{3/2}\int_{-3/2}^{3/2} f(x)dx = \frac{2}{3/2}\int_0^{3/2} f(x)dx,$$

$$= \frac{4}{3}\int_0^{3/2}\left(1-\frac{4x}{3}\right)dx = 0,$$

$$a_n = \frac{1}{3/2}\int_{-3/2}^{3/2} f(x)\cos\left(\frac{n\pi x}{3/2}\right)dx,$$

$$= \frac{2}{3/2}\int_0^{3/2} f(x)\cos\left(\frac{2n\pi x}{3}\right)dx,$$

$$= \frac{4}{3}\left(1-\frac{4x}{3}\right)\left(\frac{\sin\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)}\right) - \left(\frac{-4}{3}\right)\left(\frac{-\cos\left(\frac{2n\pi x}{3}\right)}{\left(\frac{2n\pi}{3}\right)^2}\right)_0^{3/2},$$

$$= \frac{4}{n^2\pi^2}\left[1-(-1)^n\right].$$

Also, $$b_n = \frac{1}{3}\int_{-3/2}^{3/2} f(x)\sin\left(\frac{n\pi x}{3/2}\right)dx = 0.$$

Thus $$f(x) = \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\left[1-(-1)^n\right]\cos\left(\frac{2n\pi x}{3}\right).$$

Putting $x = 0$, $$f(0) = \frac{4}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\left[1-(-1)^n\right],$$

or $1 = \dfrac{8}{\pi^2}\left[1+\dfrac{1}{3^2}+\dfrac{1}{5^2}+\ldots\ldots\right]$. Thus, $\dfrac{\pi^2}{8} = 1 + \dfrac{1}{3^2} + \dfrac{1}{5^2} + \ldots\ldots$

Exercise:

Obtain the Fourier series of the following functions over the specified intervals :

1. $f(x) = x + x^2$ over $(-\pi, \pi)$.

Answer:

$$f(x)=\frac{\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx.+2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx.$$

2. $f(x) = \left(\frac{\pi - x}{2}\right)^2$ over $(0, 2\pi)$.

Answer:

$$f(x)=\frac{\pi^2}{6}+\sum_{n=1}^{\infty}\frac{1}{n^2}\cos nx.$$

3. $f(x) = x$ over $(-\pi, \pi)$; Deduce that $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-.....\infty$.

Answer:

$$f(x)=2\sum_{n=1}^{\infty}\frac{1}{n}(-1)^{n+1}\sin nx.$$

Deduction at $x=\frac{\pi}{2}$.

4. $f(x) = |x|$ over $(-\pi, \pi)$; Deduce that $\frac{\pi^2}{8}=\frac{1}{1^2}+\frac{1}{3^2}+.....\infty$.

Answer:

$$f(x)=\frac{\pi}{2}+2\sum_{n=1}^{\infty}\frac{1}{\pi n^2}\left[(-1)^n-1\right]\cos nx.$$

Deduction at $x=\pi$.

5. $f(x) = x \sin x$ over $0 \le x \le 2\pi$. Deduce that

$$\sum_{n=2}^{\infty}\frac{1}{n^2-1}=\frac{3}{4}.$$

6. $f(x) = \begin{cases}0, & -2\le x\le 0\\ 1, & 0<x\le 2\end{cases}$ over $(-2, 2)$.

Answer:

$$f(x)=\frac{1}{2}+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)}\sin\left[\frac{(2n-1)\pi x}{2}\right].$$

7. $f(x) = x(2-x)$ over $(0,3)$.

Answer:

$$f(x)=-\frac{9}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\cos\left(\frac{2n\pi x}{3}\right)+\frac{3}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{2n\pi x}{3}\right).$$

8. $f(x) = x$ over $(-l, l)$.

Answer:

$$f(x) = \frac{2l}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}(-1)^{n+1}\sin\left(\frac{n\pi x}{l}\right).$$

9. $f(x) = \begin{cases} \pi x, & 0 \le x \le 1 \\ \pi(2-x), & 1 \le x \le 2. \end{cases}$ over the interval (0,2). Deduce that $\frac{\pi^2}{8} = \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}$.

Answer:

$$f(x) = \frac{\pi}{2} + 2\sum_{n=1}^{\infty}\frac{1}{\pi n^2}\left[(-1)^n - 1\right]\cos nx .$$

Deduction at $x = 1$.

Half-range Fourier series:

The Fourier expansion of the periodic function $f(x)$ of period $2l$ may contain both sine and cosine terms. If it is required to obtain the Fourier expansion of $f(x)$ in the interval $(0,l)$ which is regarded as half interval. The definition can be extended to the other half in such a manner that the function becomes even or odd. This will result in cosine series or sine series only.

Sine series:

Suppose $f(x) = \varphi(x)$ is given in the interval $(0,l)$. Then defining $f(x) = -\varphi(-x)$ in $(-l,0)$. Hence $f(x)$ becomes an odd function in $(-l, l)$. The Fourier series then is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \qquad (11)$$

where $b_n = \frac{2}{l}\int_0^l f(x)\sin\left(\frac{n\pi x}{l}\right)dx.$

The series (11) is called half-range sine series over $(0, l)$.

Putting $l = \pi$ in (11), the half-range sine series of $f(x)$ over $(0,\pi)$ is obtained, given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \quad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx.$$

Cosine series: Let us define

$$f(x) = \begin{cases} \phi(x) & \text{in } (0,l) \quad \text{.....given} \\ \phi(-x) & \text{in } (-l,0) \quad \text{.....in order to make the function even.} \end{cases}$$

Then the Fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right), \qquad (12)$$

$$a_0 = \frac{2}{l}\int_0^l f(x)dx,$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx.$$

The series (12) is called half-range cosine series over (0, l).
Putting $l = \pi$ in (12),

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx,$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx,$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx dx \quad \text{n} = 1,2,3,\ldots\ldots.$$

1. Expand $f(x) = x(\pi - x)$ as half-range sine series over the interval $(0,\pi)$.
Solution:

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx,$$

$$= \frac{2}{\pi}\int_0^{\pi} (\pi x - x^2)\sin nx dx,$$

Integrating by parts,

$$b_n = \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{-\cos nx}{n}\right) - (\pi - 2x)\left(\frac{-\sin nx}{n^2}\right) + (-2)\left(\frac{\cos nx}{n^3}\right)\right]_0^{\pi},$$

$$= \frac{4}{n^3\pi}\left[1 - (-1)^n\right].$$

The sine series of $f(x)$ is

$$f(x) = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^3}\left[1 - (-1)^n\right]\sin nx.$$

2. Obtain the cosine series of

$$f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases} \quad \text{over } (0,\pi).$$

Solution: Here

$$a_0 = \frac{2}{\pi}\left[\int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi}(\pi - x)dx\right] = \frac{\pi}{2}.$$

$$a_n = \frac{2}{\pi}\left[\int_0^{\pi/2} x\cos nx\,dx + \int_{\pi/2}^{\pi}(\pi - x)\cos nx\,dx\right].$$

Performing integration by parts and simplifying,

$$a_n = -\frac{2}{n^2\pi}\left[1 + (-1)^n - 2\cos\left(\frac{n\pi}{2}\right)\right],$$

$$= -\frac{8}{n^2\pi}, \qquad n = 2, 6, 10, ...$$

Thus, the Fourier cosine series is

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left[\frac{\cos 2x}{1^2} + \frac{\cos 6x}{3^2} + \frac{\cos 10x}{5^2} + ...\infty\right].$$

3. Obtain the half-range cosine series of $f(x) = c - x$ in $0 < x < c$.

Solution: Here

$$a_0 = \frac{2}{c}\int_0^c (c - x)dx = c,$$

$$a_n = \frac{2}{c}\int_0^c (c - x)\cos\left(\frac{n\pi x}{c}\right)dx,$$ integrating by parts and simplifying,

$$a_n = \frac{2c}{n^2\pi^2}\left[1 - (-1)^n\right].$$

The cosine series is given by

$$f(x) = \frac{c}{2} + \frac{2c}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\left[1 - (-1)^n\right]\cos\left(\frac{n\pi x}{c}\right).$$

Exercise:

Obtain the half-range sine series of the following functions over the specified intervals :

1. $f(x) = \cos x$ over $(0, \pi)$.

Answer:

$$f(x) = \frac{2}{\pi}\sum_{n=2}^{\infty}\frac{n}{(n^2 - 1)}\left[1 + (-1)^n\right]\sin nx.$$

2. The temperature distribution $T(x)$ at a distance x measured from one end along a metal bar of length π is given by $T(x) = x\sin x$ over $(0,\pi)$. Express $T(x)$ as a half range Fourier sine series.

Answer:

$$f(x) = \frac{2}{\pi}\sum_{n=2}^{\infty}\frac{n}{(n^2 - 1)}\left[1 + (-1)^n\right]\sin nx.$$

3. $f(x) = lx\text{-}x^2$ over $(0\ ,\ l)$

Answer:

$$f(x) = \frac{4l^2}{\pi^3}\sum_{n=2}^{\infty}\frac{\left[1-(-1)^n\right]}{n^3}\sin\left(\frac{n\pi}{l}\right)x.$$

Obtain the half-range cosine series of the following functions over the specified intervals :

4. $f(x) = x^2$ over $(0,\pi)$

Answer:

$$f(x) = \frac{\pi^2}{3} - 4\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n^2}\cos nx.$$

5. $f(x) = (x\text{-}1)^2$ over $(0,1)$.

Answer:

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2}\sum_{n=2}^{\infty}\frac{1}{n^2}\cos n\pi x.$$

Obtain the half-range cosine and sine series of the following functions over the specified intervals :

6. $f(x) = \begin{cases} kx, & 0 \le x \le \dfrac{l}{2} \\ k(l-x), & \dfrac{l}{2} \le x \le l. \end{cases}$

Answer:

Cosine series:

$$f(x) = \frac{kl}{4} + \frac{2kl}{\pi^2}\sum_{n=2}^{\infty}\frac{\left[2\cos\left(\frac{n\pi}{2}\right)-1-(-1)^n\right]}{n^2}\cos\left(\frac{n\pi}{l}\right)x.$$

Sine series:

$$f(x) = \frac{4kl}{\pi^2}\sum_{n=2}^{\infty}\frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)\sin\left(\frac{n\pi}{l}\right)x.$$